

EtherCAT SubDevice Errata for PRU ICSS EtherCAT Firmware Version x.5.46

PD/PDI watchdog counter issue

- **Issue/ Failure Description or state**

- SDOCM00098105/PINDSW-72: PDI/PD watchdog counter incremented by 1 whenever PDI/PD watchdog is disabled using EtherCAT master

- **Conditions in which failures occur**

- Whenever EtherCAT master disables WD by writing zero to respective Watchdog Time registers (0x410:0x411 or 0x420:0x421)

- **Root cause**

- This is PRU-ICSS h/w behavior

LRD access on unused registers

- **Issue/ Failure Description or state**
 - SDOCM00098950/PINDSW-74: LRD access on unused registers results in WKC increment
- **Conditions in which failures occur**
 - LRD access on unused registers result in WKC increment
- **Root cause**
 - Firmware does not support register protection in LRD mode at this moment, it requires more firmware footprint to support, this minor spec compliance does not justify the footprint increase and no Write Only registers in ESC

Error frames with “SOF but no SFD” are not counted if arrived on reverse path port

- **Issue/ Failure Description or state**

- PINDSW-2204 : If a frame with SOF but no SFD arrives on the reverse path, EtherCAT SubDevice cannot detect such frame. Therefore, it also doesn't count this frame in the error counters. According to EtherCAT specs it should be counted. These error frames are detected and counted on the forward path. The issue is only for the reverse path.

- **Conditions in which failures occur**

- With at least two devices connected in the chain, on the device with both ports connected, if a frame without SFD arrives on the OUT port (reverse path port), then that frame is not counted in any of the error counters of the ESC
- When the hardware forwards these frames (without SFD) it adds a SFD and a specific pattern "0x5a 0x5a 0x5a 0x5a" at the end of the frame along with an odd nibble

- **Root cause**

- Because of the way firmware handles the Ethernet frames on reverse path, there is no hardware/software provision for the firmware to even detect frames without SFD. Thus, it cannot increment the error counters accordingly.

- **Work around**

- When such frame (without SFD, very rare phenomenon) is received by a SubDevice, even though the receiving SubDevice cannot detect the frame, it forwards the frame with added odd nibble. So all the following SubDevices will detect that frame as forwarded error.
- Also when the SubDevices forwards the frame, it adds a specific pattern "0x5a 0x5a 0x5a 0x5a" at the end of the frame, which can be identified using wireshark logs.

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NOTE: This issue will not be fixed

System time of next Sync0 pulse register (0x990:0x993) is not updated instantaneously

- **Issue/ Failure Description or state**

- PINDSW-2360: System time of next Sync0 pulse register (0x990:0x993) is not updated instantaneously, resulting in read of incorrect value if read immediately after sync pulse.

- **Conditions in which failures occur**

- When Sync0 pulse register is read immediately after a sync pulse occurs.

- **Root cause**

- The Sync0 pulse register is updated in the firmware with a typical delay of ~1.7us and worst case delay of ~5.6us. If the Sync0 pulse register is read after the sync pulse but before the register is updated (before the above limit expires) it can result into read value returning previous pulse time which is incorrect.

Read permissions and byte level write permissions for RW type commands are not checked

- **Issue/ Failure Description or state**

- PINDSW-5135: Read permissions and byte level write permissions for RW type commands are not checked. Write permissions will be checked at word level.

- **Conditions in which failures occur**

- When the firmware processes APRW, FPRW, BRW and LRW commands

- **Root cause**

- The firmware process path for this particular scenario is taking longer time to process if byte level read and write permission checks are present. This leads to firmware not able to process the command correctly and therefore failing the RW operation.

RX_ER counter does not count errors outside frame in MII RX_CLK units precisely

- **Issue/ Failure Description or state**

- PINDSW-5145: RX_ER counter does not count errors outside frame in MII RX_CLK units precisely

- **Conditions in which failures occur**

- When the RX_ER error occurs

- **Root cause**

- Firmware counts RX_ER only once per frame, i.e. only when RX_DV is asserted.
- For any RX_ER seen when RX_DV is not asserted, firmware does not increment this counter.

- **Work around**

- Added PHY RX Error Counter Register (0x0E28) for improving RX Error Counter accuracy. Configuring this register will track RX_ERs within a frame precisely using PHY registers.
- Refer the “Register Exceptions -> RX Error Counter of Port 0 & 1 (0x0301 and 0x0303)” section in “Industrial Communications Toolkit->Third Party Protocol Stacks or Customers using their own Stacks->EtherCAT SubDevice FWHAL->TI EtherCAT SubDevice Controller Exceptions” page of SDK documentation for more details.

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NOTE: This issue will not be fixed, workaround is available.

Link Lost Counter is incorrectly incremented once for ports with polarity of RXLINK input as “Active Low” during initialization

- **Issue/ Failure Description or state**

- PINDSW-5414: Link Lost Counter is incorrectly incremented once for ports with polarity of RXLINK input as “Active Low” during initialization

- **Conditions in which failures occur**

- When the polarity of RXLINK input is “Active Low”, with MLINK mode of link detection enabled.
- It is incremented incorrectly only during MDIO initialization when setting MLINK mode in MDIOUSERPHYSEL0/MDIOUSERPHYSEL1 register.

- **Root cause**

- In case of “Active Low” polarity, there is a change in MDIOLINK register when the link detection mode is switched to MLINK mode in MDIO. As without an active link, pin state will be held HIGH but from MDIO IP point of view when MLINK mode is enabled, there is a transition of MDIOLINK STATE from LOW to HIGH.
- This change leads to MDIO Interrupt which triggers a link change event in PRU-ICSS INTC, which increments the Link Lost Counter Register (0x310).

- **Work around**

- Use Active High Polarity for LED_LINK/SPEED connected to MII0/MII1 Receive Link (RXLINK) pin of PRU-ICSS

Issues fixed in PRU-ICSS-EtherCAT Firmware Version x.5.46

Industrial Communications SDK v10.00

Process Data buffer swap not swapping in FreeRun mode

- **Issue/ Failure Description or state**

- PINDSW-8246: Fix the FreeRun mode PDI side write buffer swap issue.

- **Conditions in which failures occur**

- When TwinCAT is configured to run in FreeRun (by writing 0 to 0x1C33.bit1), the process data address seems to swap (triple buffer implementation) but after certain intervals of time, it is not swapping among the 3 buffers as expected.

- **Root cause**

- Mismatch in swapping related code within the firmware.

- **Fix:**

- Fixed the mismatch and tested for the same.

LRW access to non-interleaved input and output process data of multiple SubDevices does not work

- **Issue/ Failure Description or state**

- SDOCM00105048/PINDSW-141: LRW access to non-interleaved input and output process data of multiple SubDevices does not work. SOEM accesses SubDevices in LRW mode this way

- **Conditions in which failures occur**

- Single LRW datagram accessing FMMU mapped areas in multiple SubDevices and PD out is mapped
 - FMMU0(0x1000:0x1007)->SM2 #1(Write SM) FMMU1(0x1008:0x100F)->SM2#2 (Write SM)
 - FMMU2(0x1010:0x1017)->SM3 #1 (Read SM) FMMU3(0x1018:0x101F)->SM3#2(Read SM)
- Single LRW access from (0x1000:101F)

- **Root cause**

- Increased code memory requirements in firmware to implement this support as well as non-interleaved access I/O data is not a very optimal use of EtherCAT – it increases the cycle time overhead/datagram size and not effective use of LRW datagram which can perform read and write in the same cycle.

- **Fix:**

- Added logic to implement this support within the firmware. Tested this using default mode of few open-source EtherCAT MainDevices like SOEM and IgH.
- Dynamic increase of TX_START_DELAY within firmware to take care of the timing requirement when this topology/feature is implemented.

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NOTE: This issue will not be fixed in AM335x due to memory limitation.

Single datagram accessing multiple FMMU mapped areas using LRD/LWR commands

- **Issue/ Failure Description or state**

- SDOCM00092510/PINDSW-47 : Single datagram accessing multiple FMMU mapped areas in a single SubDevice will only update the data corresponding to first FMMU in datagram

- **Conditions in which failures occur**

- Single datagram accessing multiple FMMU mapped areas in single SubDevice.

FMMU0(0x1000:0x1007)->SM2 (Write SM)

FMMU1(0x1000:0x1007)->SM3 (Read SM)

FMMU2(0x1008:0x100F)->SM4 (Write SM)

FMMU3(0x1008:0x100F)->SM5 (Read SM)

- Single LRD to access (0x1000:100F) will only access SM3
- Single LWR to access (0x1000:100F) will only access SM2
- 2 LRD/LWRs are required in this case to access both pair of SMs - LRD1/LWR1(0x1000:0x1007) and LRD2/LWR2(0x1008:0x100F)

- **Root cause**

- Increased code memory requirements in firmware to implement this support

- **Fix:**

- Added logic to implement this support within the firmware. Tested this using TwinCAT by writing and reading from multiple FMMU mapped areas.
- Dynamic increase of TX_START_DELAY to take care of the timing requirement when this topology/feature is implemented.

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NOTE: This issue will not be fixed in AM335x due to memory limitation.

Issues fixed in PRU-ICSS-EtherCAT Firmware Version x.5.20

Industrial Communications SDK v09.02

Drift Compensation issue for filter values other than 0

- **Issue/ Failure Description or state**

- PINDSW-7521: Drift Compensation issue for filter values (ESC Reg.0x0934) other than 0

- **Conditions in which failures occur**

- When checking ESC register 0x92C:System Time Difference, the static drift compensation is not converging for different values configured in ESC Reg.0x0934.

- **Root cause**

- Memory corruption while calculating mean and related calculations for the drift compensation as the same memory space was being used by the different core which was corrupting the drift compensation.

PR0_IEP0_EDIO_DATA_IN_OUT30 not functioning as AL event

- **Issue/ Failure Description or state**

- PINDSW-7492: "PR0_IEP0_EDIO_DATA_IN_OUT30" is not functioning as AL event.

- **Conditions in which failures occur**

- IEP pin register value for PR0_IEP0_EDIO_DATA_IN_OUT30 was incorrect.

- **Root cause**

- Configure the correct value for PR0_IEP0_EDIO_DATA_IN_OUT30

Issues fixed in PRU-ICSS-EtherCAT Firmware Version x.5.18

Industrial Communications SDK v09.01

Workaround for HW errata with 64-bit IEP timer read from PRU

- **Issue/ Failure Description or state**

- Reading the 64-bit IEP timer does not have lock MSW (Most significant word) logic when LSW (Least significant word) is read and this can lead to issues when 32-bit LSW wraps around, there were no visible failures with EtherCAT DC mode due to this

- **Conditions in which failures occur**

- The issue is seen when IEP counter (IEP_COUNT_REG1 : IEP_COUNT_REG0) is read from ICSS PRU cores as 64-bit read operation.

- **Root cause**

- Identified by the HW team during debug of an unrelated issue, PRU needs to check LSW value and read the IEP again when LSW is 1 clock cycle behind 0xFFFF_FFFF to 0 transition

ESC Reg.0x800+5*SM_index.bit1 does not change on PDI side mailbox read

- **Issue/ Failure Description or state**

- PINDSW-7099 : ESC Reg. 0x800+5*SM_index.bit1 remains 0 and does not change on PDI side mailbox read.

- **Conditions in which failures occur**

- Status bit is not set after PDI reading completely and successfully the mailbox buffer.

- **Root cause**

- This was a PRU firmware bug, firmware was not setting this bit and fixed. Workaround will be to poll mailbox empty status from the EtherCAT master side

Sync0/1 Out missing very rarely during frequent PREOP to OP transition tests

- **Issue/ Failure Description or state**

- PINDSW-7017 : AM263x: Sync0/1 Out missing very rarely during EtherCAT state machine change from PREOP → SAFEOP → OPERATIONAL.

- **Conditions in which failures occur**

1. Happen once every many times of the EtherCAT state machine change from PREOP → SAFEOP → OPERATIONAL.
2. Once the EtherCAT master configures the registers and the timing registers correctly the PRU does not toggle the Sync0/1 pin at all.
3. Re starting the EtherCAT state machine, i.e. PREOP → SAFEOP → OPERATIONAL, Sync0/1 generation recovers to normal behavior.

- **Root cause**

- Identified a PRU firmware race condition w.r.t CMP1 HIT already seen but SYNC0 pulse has not yet started scenario, this caused firmware to miss subsequent sync generation portion.

Issues fixed in PRU-ICSS-EtherCAT Firmware Version x.5.12

MCU+ SDK v08.05

ESC DL Status register is not initialized correctly

- **Issue/ Failure Description or state**
 - PINDSW-5384 : ESC DL Status register is not initialized correctly
- **Conditions in which failures occur**
 - Initial value of these registers is not correct. Port 1 loop is open, even if there is no link up
- **Root cause**
 - Ports should be closed by default

Next SYNC1 Pulse register is not updated correctly

- **Issue/ Failure Description or state**

- PINDSW-5385 : Next SYNC1 Pulse register is not updated correctly

- **Conditions in which failures occur**

- Value in this register is not consistent with the SYNC1 pulse timing

- **Root cause**

- SYNC1 Pulse was being generated one SYNC0 cycle late (PINDSW-5401)
- No value was being set in Next SYNC1 Pulse register (0x0998:0x099F) until the first SYNC1 pulse occurs

SYNC1 Pulse is generated one SYNC0 cycle time late

- **Issue/ Failure Description or state**
 - PINDSW-5401 : SYNC1 Pulse is generated one SYNC0 cycle time late
- **Conditions in which failures occur**
 - When SYNC1 is enabled
- **Root cause**
 - First SYNC1 pulse was being triggered one cycle late than expected

Issues fixed in PRU-ICSS-EtherCAT Firmware Version x.5.8

MCU+ SDK v08.04

Write to System Time Delay (0x0928) register works only once

- **Issue/ Failure Description or state**

- PINDSW-5369 : Write to System Time Delay (0x0928) register works only once

- **Conditions in which failures occur**

- Second write to 0x0928 register does not take effect

- **Root cause**

- The value was being read from the register only once in the beginning. Therefore no subsequent updates to this register work.

Issues fixed in PRU-ICSS-EtherCAT Firmware Version x.5.0

MCU+ SDK v08.02

Lost Link Counter register (0x310/0x311) increments with "2" on every link down event instead of "1"

- **Issue/ Failure Description or state**

- PINDSW-3120 : The Lost Link Counter register (0x310 and 0x311) increments with "2" on every link down event instead of "1"

- **Conditions in which failures occur**

- On Port0/1 Link disconnect. The Lost Link Counter register counts the link down events on the Port.

- **Root cause**

- Set the type of MDIO Link interrupt to Edge in tiesc_pruss_intc_mapping.h file
- The reason for this issue is the MDIO link interrupt was mapped to PRU as PULSE interrupt. The Pulse nature of MDIO Link interrupt lead to PRU interrupt being set twice. Therefore, the firmware increments the counter twice for every link down event.
- This change to Pulse mode was done to fix link stability issue in a certain rare condition after multiple link disconnect-connect cycles. The issue led to the Port always remaining closed until we have link disconnect/connect action on the other port.

DC timings are not correct after re-activation of DC

- **Issue/ Failure Description or state**
 - PINDSW-5194 : DC timings are not correct after re-activation of DC
- **Conditions in which failures occur**
 - When DC mode is reactivated, SYNC1 generation is not correct. For example, if SYNC1 is programmed to trigger once per 8 cycles of SYNC0 and before DC was deactivated, there were 4 cycles of SYNC0 after last SYNC1 pulse, then after activating DC, SYNC1 is seen after 4 SYNC0 cycles.
- **Root cause**
 - Internal counters were not being cleared

ESC firmware counts short frames in 0x0302 register

- **Issue/ Failure Description or state**
 - PINDSW-5229 : ESC firmware counts short frames in 0x0302 register
- **Conditions in which failures occur**
 - When both ports are connected and short frame arrives on Port 1
- **Root cause**
 - 0x0302 error counter is being incremented by firmware for short frames. Only 0x030C should be incremented.

Lost Link Counter register (0x310/0x311) is getting initialized to 1

- **Issue/ Failure Description or state**
 - PINDSW-5267 : Lost Link Counter register (0x310/0x311) is getting initialized to 1
- **Conditions in which failures occur**
 - During power up, this counter is updated to 1 even without a physical link loss.
- **Root cause**
 - It is being initialized to 1 by default, even when there is no link drop.

Issues fixed in PRU-ICSS-EtherCAT Firmware Version 5.4.243

RW type commands for address 0x300 and length 0x20 fails

- **Issue/ Failure Description or state**

- PINDSW-4989 : EtherCAT SubDevice does not update the WKC if a RW command is received for address 0x300 with length of 0x20

- **Conditions in which failures occur**

- Master sends and FPRW command for address 0x300 and length 0x20

- **Root cause**

- The firmware process path for this particular scenario is taking longer time to process. This leads to firmware not able to process the command correctly and therefore failing the RW operation.
- If read permission checks and byte level write permission checks are skipped, firmware is able to process the commands. Write permissions will still be checked at word level. See PINDSW-5135 for more details.